

Education

University of Illinois at Urbana-Champaign

Aug 2025 — May 2028

B.S. in Mathematics and Computer Science (Technical GPA: 4.0, Overall GPA: 3.99); Honors: James Scholar, Dean's List x2

- Relevant Coursework: Systems Programming, Computer Architecture, Data Structures, Algorithms, Distributed Systems

Technical Skills and Awards

Awards and Recognitions: YC Startup School 2026, Keywords AI Hackathon 2025 Winner

Languages: Lean4, Dafny, Python, C++, C, Go, Verilog, Rust, SQL, React, TypeScript, HTML/CSS/JS

Tools & Libraries: Linux, Tensorflow, NumPy, MongoDB, FastAPI, AWS Bedrock, Claude Code/Codex/Cursor

Research

FOCAL Lab, Formal Verification & Verified LLM Decoding Research

Sept 2025 — Present

- Building **MetaDecode**, a verified constrained-decoding synthesis pipeline comprising **2,700+** lines of **Dafny** with **60+** cost-bounded CSD helpers whose pre/postcondition contracts gate every LLM-synthesized strategy before **Python** transpilation.
- Architected unified baseline adapters, stratified evaluation splits, and syntax-annotated scoring, benchmarking **100+** configurations against expert baselines to calibrate **MetaDecode** acceptance and early-stopping criteria.
- Implemented generation, verification, compilation, and evaluation of CSDs across **GSM-Symbolic**, **Spider**, and **SMILES** benchmarks, running ablation studies over **Qwen3.5** evaluators (**2B/4B/9B**) with held-out comparisons against CRANE, IterGen, and CARS.

Experience

University of Illinois at Urbana-Champaign, Systems Programming Course Assistant

Aug 2026 — Present

- Leading weekly lab sessions for **30+** students covering **processes, threads, synchronization, and memory management** in C on Linux.
- Co-developing the **midterm and final exams** and editing the course reference book, aligning assessment content with systems programming learning objectives.
- Guiding students one-on-one through debugging C implementations of **memory allocators, schedulers, TCP servers, and filesystems** during office hours.

AbbVie, Generative AI Platform & Workflow Automation Intern

June 2024 — Aug 2024

- Built a **Streamlit** interface for CRUD management of **300+** JSON configuration files powering an internal Generative AI API, and migrated **300+** prompt templates into **PostgreSQL**.
- Designed a modular configuration system enabling dynamic pipeline updates without code changes, reducing reconfiguration time from **hours to minutes** for internal AI workflows.

Technical Projects

Two-Level Segregated Fit Malloc, High-Performance Memory Allocation

- Developed a custom memory allocator (malloc, calloc, realloc, free) in C, outperforming glibc by up to **100%** in throughput with over **300%** memory reduction on targeted benchmarks, placing **3rd of 300** students.
- Designed a **two-level segregated fit** (TLSF) allocator with **128-bit** bitmask free-list indexing, localized arena allocation, and a deferred-free cache to exploit repeated alloc/free patterns.
- Engineered aggressive block-level optimizations including compressed **32-bit** pointers, boundary-tag coalescing with footer elision, and flag-encoded headers for **O(1)** neighbor status checks.

Pragma, Provider-Agnostic Agentic Coding Framework

- Built a terminal-based agentic coding framework in **Go** with a multi-provider LLM adapter supporting **OpenAI**, **Anthropic**, and **OpenRouter**, featuring native and text-based tool calling with automatic provider detection.
- Designed a cost budgeting system with per-task token tracking, a pricing table with fuzzy model matching, and automatic model tiering that cascades across providers, downgrading to cheaper models as budget thresholds are hit.
- Engineered an event-driven async process manager with disk-backed output buffering, timeout enforcement, platform-agnostic process tree killing, and output filtering that extracts compiler errors from large build logs to minimize token usage.

- Developing a **SQLite-backed knowledge graph** with structural indexing, temporal versioning, and behavioral analysis layers to power **graph-guided task decomposition** via co-change clusters, with a sequential plan architecture supporting topological-sort parallelism through `depends_on` metadata.
- Implemented auto-generated design docs and a living architecture document with diff-based incremental updates, providing persistent session memory across agent restarts.

Squishy.ai, *LLM-Driven Soft Robotics Simulation Platform*

- Built a full-stack platform that translates natural language into executable soft robotics physics simulations via **PyElastica** (Cosserat rod theory), placing as a **winner** at the Keywords AI hackathon among **200+** participants.
- Architected a high-performance **FastAPI** backend with **Numba** JIT compilation and a headless **Matplotlib/FFmpeg** rendering pipeline for end-to-end simulation generation in **under 20 seconds**.
- Developed an interactive **React/Vite** frontend featuring real-time simulation visualization and LLM-routed code generation through the Keywords AI gateway.

Abstract Algebra, the Game, *Interactive Lean4 Proof Game*

- Developed an interactive **Lean4** proof game hosted on the Lean Game Server comprising **2 worlds** and **24 levels** that teach group theory through guided theorem proving, covering abelian groups, subgroups, and normal subgroups.
- Formalized **1,200+** lines of **Lean4** encoding identity uniqueness, inverses, cancellation laws, commutators, conjugation, subgroup intersection, and the one-step subgroup test, building a notation-agnostic `MyGroup` typeclass.
- Designed progressive level sequences that decompose **formalized proofs** into **structured learning blocks**, with layered hints and narrative annotations guiding players from basic group axioms to advanced subgroup characterizations.

Asynchronous Audio Relay, *Low-Latency Concurrent Systems & Networking*

- Architected a multithreaded UDP store-and-forward audio relay in **C** (Linux/Raspberry Pi) with **POSIX sockets** and per-peer **pthreads**, replacing a single-threaded poll loop that serialized concurrent sessions and amplified head-of-line blocking.
- Validated **<500 ms** end-to-end latency on live two-party calls across a **900 mile** network path, demonstrating that store-and-forward relay semantics need not trade away interactive responsiveness.
- Engineered a **lock-free** ring buffer and append-only audio store for concurrent ingest and asynchronous voicemail playback, eliminating mutex contention on the hot path versus a mutex-backed prototype that introduced measurable tail latency under load.

Logana, *Privacy-First Encrypted Messaging Platform*

- Architecting a **C++20** end-to-end encrypted group messaging platform with lock-free **SPSC chunked deque** storage, tag-based message routing, and permission-gated fan-out over a relay architecture.
- Implemented cross-platform async TCP transports with **io_uring** on Linux and **IOCP** on Windows, unified under a platform-abstracted event loop with a typed alias system and forward-compatible framing protocol.
- Designed a store-and-ref pipeline where the relay router performs permission checks, stores messages in per-account chunked deque containers, and fans out lightweight **MessageRefs** to connected devices.

Cherimoya, *Systems Language Compiler Frontend*

- Designing a **C++20** compiler frontend for Cherimoya, a systems language featuring ownership-oriented primitives (`val`, `addr`, `move`, `free`) and a **15-level** operator precedence grammar, targeting a **self-hosting compiler** within a ground-up personal computing stack.
- Built a Pratt (precedence-climbing) expression parser over a `unique_ptr` AST with nud/led binding powers, left/right associativity, and `ErrorExpr`-preserving recovery for multi-error reports.
- Implemented a span-based lexer with maximal-munch scanning, offset-native source locations, and multi-error recovery, covered by **Catch2** tests across operators, numeric bases, and keywords.
- Extending the parser with statement and declaration AST nodes, type parsing with qualifier ordering enforcement, three repeat forms, two pattern-match modes, and **contract clause** parsing for function-level pre/postconditions.

SIGecom, *Systematic Trading Strategy Research & Simulation*

- Designed and backtested **multi-asset algorithmic trading strategies** in **Python** combining SMA crossovers, EMA+RSI filters, MACD momentum, and statistical arbitrage across **6 symbols**.
- Achieved **net-positive PnL** across a diversified portfolio spanning cryptocurrency (BTC, ETH, SOL) and foreign exchange (INR, ZAR, JPY) markets.
- Deployed strategies into a live simulation framework executing **40+ trades** over a one-week evaluation period on real-time cryptocurrency and FX market data.

Financial Engineering, *Quantitative Options Pricing & ML-Based Volatility Modeling*

- Applied **neural network de-Americanization** to convert American option prices into European equivalents, enabling accurate **implied volatility surface** construction via Black-Scholes inversion.
- Implemented the **Cox-Ross-Rubinstein** binomial lattice in **Python** to price European and American options, validating convergence to Black-Scholes analytical values across strike and maturity grids.
- Characterized **early exercise boundaries** across varying dividend, volatility, and interest rate regimes, quantifying the accuracy-performance tradeoff as a function of lattice depth.